



Neurological and neurodevelopmental manifestations in children and adolescents with Down syndrome

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Abstract

Down syndrome (DS) is a common genetic condition that is associated with a wide range of medical and developmental challenges, including neurological and neurodevelopmental conditions. DS is caused by three copies of chromosome 21 and is the most common genetic cause of intellectual disability (ID). Beyond ID, individuals with DS may also manifest many characteristic medical, neurological, neurobehavioral, and developmental conditions that are complex and result in a variety of clinical presentations with varying etiologies, diagnostic strategies, and interventions. In this clinical review, we present an overview of neurobiology in DS and potential pharmacological targets that may impact the characteristic developmental, cognitive, and learning profile seen in DS. We examine medical conditions that may impact development and functioning as well as neurologic findings such as hypotonia and motor impairments, epilepsy, and abnormal movements. Finally, we extensively discuss the nuances of behavioral, neurodevelopmental, and mental health conditions that may co-occur in children and adolescents with DS and may sometimes be misattributed to manifestations of ID alone. This chapter seeks to draw attention to these conditions that necessitate keen diagnostic discernment, specialized evaluation and treatment, and further research to better inform clinical care.



1. Introduction

Down syndrome (DS) is the most common genetically acquired cause of intellectual disability (ID) with an estimated prevalence of 1 in 700 live births in the United States (Mai et al., 2019). DS is characterized by a constellation of physical and medical features first identified as a syndrome by John Langdon Down over 200 years ago (Down, 1866). However, it was

Jerome Lejeune's 1953 scientific discovery that individuals with phenotypic DS had triplication of all or part of chromosome that has paved the way toward a better biological understanding of this condition (Roizen & Patterson, 2003; Lejeune, Gautier, & Turpin, 1959). The extra chromosomal material from the triplicated 21st chromosome results in alterations in brain functioning leading to nearly universal intellectual disability in addition to higher rates of other neurologic, neurodevelopmental, and mental health conditions, including autism, attention deficit hyperactivity disorder (ADHD), anxiety, and others (Capone, Goyal, Ares, & Lannigan, 2006; Dykens, 2007; Palumbo & McDougle, 2018). Additionally, the extra 21st chromosome can affect almost all body organ systems such that individuals with DS may develop a wide range of medical conditions (Bull, 2011, 2020).

When thinking broadly about intellectual disability as a categorical neurodevelopmental disability, certain common characteristics are commonly addressed in clinical care. Here, we specifically focus on Down syndrome, which has a unique clinical presentation with specific features common to the syndrome in addition to substantial heterogeneity. Research advances have revealed a better understanding of neurobiology, developmental and behavioral phenotype, medical conditions and co-occurring neurological, neurodevelopmental and mental health conditions. The interplay between these areas is not yet well understood and is an important topic for future investigation. This review will focus on neurodevelopmental manifestations in children and adolescents with DS in the context of what we know about genetics, neurobiology, neuropsychologic functioning, and the larger milieu of medical conditions that may impact brain development and functioning.

Within the past several decades, the scientific community has made strides in understanding the connection between genetics, cellular biology, and human physical and cognitive development, helping to improve our grasp of how the extra material from chromosome 21 may contribute toward the DS phenotype. From a neurodevelopmental standpoint, the impact of trisomy 21 begins during the fetal period and lasts throughout the lifespan, leading to decreased formation and growth of neurons, altered connectivity throughout the brain, atypical signal transduction, and a propensity toward neurodegeneration and neuroinflammation. These aberrations impair certain aspects of brain functioning, leading to a characteristic neurobehavioral profile that will be presented throughout this review. We will further examine what we know about the neurobiology of DS and how these neurobiological differences may impact neurodevelopmental outcomes.

Neurobehavioral and cognitive phenotypic characterization in DS has outpaced our neurobiological understanding of DS and is fairly well-developed, though many questions remain. The phenotype includes global delays with more impairments in early motor and expressive language skills and long-standing language and communication challenges that tend to persist throughout life. Universal intellectual disability occurs with particular challenges with executive functioning and memory, though a broad range of abilities and functioning seen among different individuals with DS. When narrowing in on cognition, many individuals have uneven skill profiles with relative strengths in nonverbal reasoning and visual processing, especially when compared to verbal and auditory processing skills. This cognitive and learning profile is distinct and commonly seen to varying degrees among individuals with DS, carrying implications for therapeutic and educational programming that will be discussed further in the review.

There is significant inter-individual variability in neurodevelopmental outcomes in DS and a relatively poor understanding of how underlying biology and other factors may predict cognitive phenotypes. Beyond the presence of an extra chromosome, the DS phenotype is certainly impacted by an individual's background genetics, environmental exposures and enrichment, and other as-of-yet undetermined factors that researchers are seeking to better understand. Because the biological underpinnings and environmental factors that may contribute to phenotypic differences are poorly understood, neurodevelopmental outcomes cannot be predicted based on genetic testing alone or other biomarkers at this time. A better understanding of the interplay between both biologic and environmental factors in DS is also critical in the quest to identify potentially modifiable factors that could one day be targets for therapeutic intervention to optimize neurodevelopmental and health outcomes. Several attempts at precision treatment of particular neurobiological targets have been unsuccessful to date but pave the way for new research directions, as discussed in the review. Currently, environmental modifications and early intensive educational and therapeutic enrichment remain mainstays of treatment to address ID and other neurobehavioral issues in DS, though responses to these interventions also show great inter-individual variability.

Individuals with DS are at higher risk for many medical co-occurring conditions which may have potential negative impacts on neurodevelopmental outcomes. This supposition largely stems from information about these health conditions and how they may impact neurodevelopmental phenotypes in the general population. For example, altered cognitive and behavioral outcomes

have been seen in such conditions as congenital heart disease, untreated hypothyroidism, epilepsy, and obstructive sleep apnea, all of which occur at higher rates among individuals with DS. It is hypothesized that these medical conditions contribute directly toward altered neurodevelopmental outcomes via consequent changes in physiology (for example, congenital heart disease may lead to poorer blood flow and oxygen delivery to certain parts of the brain, thereby causing more selective dysfunction in those areas, leading to particular cognitive deficits). Additionally, neurodevelopmental differences may also come about as the result of a shared genetic or epigenetic mechanisms between health and cognitive phenotypes, though this concept is poorly understood (Startin et al., 2020). The associations between medical and neurodevelopmental conditions in DS are reviewed here and require further investigation to determine optimal management throughout the lifespan.

Finally, this review will discuss the milieu of neurologic, neurodevelopmental, behavioral, and mental conditions seen among children with Down syndrome, all of which often impact outcomes and quality of life in more significant ways than cognitive functioning alone. Conditions such as hypotonia and motor impairments, epilepsy, and abnormal movements are discussed. We cover the nuances of behavioral, neurodevelopmental, and mental health conditions that may co-occur in children and adolescents with DS and may sometimes be misattributed to manifestations of ID alone. Conditions such as autism (ASD), attention-deficit hyperactivity disorder (ADHD), challenging behaviors, and sensory processing differences will be examined thoroughly, including details about presentation, diagnosis, and management. Additionally, mental health conditions such as depression, anxiety, and unexplained regression are discussed.

Because neurologic, neurobehavioral, and mental health conditions require distinctive management approaches or therapeutic interventions that might not otherwise be available, it is critical that children with DS have access to thorough screenings and prompt diagnostic evaluations for co-occurring neurodevelopmental and behavioral conditions. The diagnostic process is complex and nuanced, often best reserved for clinicians with extensive experience working with individuals with DS. However, an increasing awareness of these co-occurring conditions among all clinicians who interact broadly with individuals with DS will undoubtedly benefit this population. Additionally, given their significant impact on everyday functioning, these neurodevelopmental, behavioral, and mental health conditions are now being recognized as critical targets for more extensive investigation. As discussed, further research is needed to guide the

standardization of clinical care approaches, to develop new and effective treatment modalities, and to support all individuals with DS in meeting their full potential.



2. Neurobiology

The human chromosome 21 (Hsa21) is the smallest autosome, containing over 500 genes, though most do not appear to be essential for normal development (Gardiner et al., 2010; Sturgeon, Le, Ahmed, & Gardiner, 2012). Several “dosage sensitive” regions have been mapped across the length of Hsa21 which have been shown to be sufficient for inducing the complete phenotype of DS in mouse models (Belichenko, Kleschevnikov, Salehi, Reeves, & Mobley, 2009; Korb et al., 2009). However, the neurodevelopmental phenotype of DS may be attributable not just to specific triplication of Hsa21 genes in isolation, but rather to over-expression or under-expression of entire downstream functional pathways (Gardiner et al., 2010).

In DS, structural differences are observed in the brain, including smaller frontal and temporal lobe volumes and an adult brain that is about 20% smaller compared to the general population (Lott, 2012; Schmidt-Sidor, Wisniewski, Shepard, & Sersen, 1990). The brain in DS reveals widened gyri and shallow sulci in much of the cortex aside from the posterior parietal and occipital cortex and relatively spared subcortical white matter (White, Alkire, & Haier, 2003; Wisniewski, 1990). The cerebellum is often small and morphologically atypical with reduced numbers of granule cells, which may contribute to hypotonia and challenges with attention, executive functioning, language learning, working memory, affect, and emotional responses (Lott, 2012; Strick, Dum, & Fiez, 2009).

This altered brain development results from decreased neurogenesis, reduced synaptic plasticity, altered neurotransmission, and other aberrant processes (Capone, 2016; Lott, 2012; Lott & Dierssen, 2010). Ultimately the brain in DS has fewer overall neurons that are also abnormal in shape and size (Becker, Mito, Takashima, & Onodera, 1991; Schmidt-Sidor et al., 1990., Larsen et al., 2008). There is reduced synaptic density and plasticity (Raz et al., 1995) and altered synaptic transmission (Marin-Padilla, 1976; Torres, Garcia, Tang, & Busciglio, 2018). This more limited and atypical neuronal connectivity ultimately impacts the ability of the brain to process information (Capone, 2016; Lott, 2012). Synaptic connections

throughout the brain are shorter, leading to more focal networks at the expense of typical interhemispheric connectivity (Lott, 2012). Because of this, cognitive functions that require processing involving networks between multiple different parts of the cortex (language, attention, social-cognition abilities) may be less efficient in DS.

Altered function of neurotransmitters is related to the above processes and also contributes to abnormal neurodevelopment, including atypical GABAergic, glutamatergic, noradrenergic, cholinergic, and serotonergic neurotransmission (Capone, 2016; Lott, 2012; Lott & Dierssen, 2010). The abnormal balance between inhibition and excitation may lead to reduced long-term potentiation and increased long term depression, reducing synaptic plasticity and the potential for learning. Neurotransmitter systems could be potential targets for pharmacologic intervention to positively augment neurodevelopment.



3. Cognitive therapeutics

Though our understanding of the neurocognitive and behavioral profiles in DS has grown significantly over time (see below), to date there are no pharmacological treatments that effectively improve the core cognitive deficits or adaptive limitations seen in DS. Cognitive therapeutic research in DS expanded in a meaningful way with the creation of the Ts65Dn mouse model, moving research from anecdotally based pharmacologic studies in the 1960–1980s toward true translational research based on identification of possible molecular targets in mice by the 1990s (Davisson, Schmidt, & Akeson, 1990; Hart et al., 2017; Reeves et al., 1995). Primary targets investigated in pre-clinical intervention studies with mouse models have included serotonin, gamma-aminobutyric acid (GABA), dual-specificity tyrosine phosphorylation-regulated tyrosine kinase 1a (DYRK1a) protein, A-beta protein, and *N*-methyl-D-aspartate (NMDA) receptors (Bartesaghi et al., 2015; Costa & Scott-McKean, 2013; Gardiner, 2015).

This research has provided the impetus for investigation of specific pharmacological interventions via human clinical trials. Thus far, clinical trials using agents such as donepezil, piracetam, rivastigmine, folinic acid, memantine, EGCG, Basmisanil, pentalyne-tetrazole, scyllo-inositol, intranasal glulisine, and fluoxetine have failed to reveal any significant differences in neurodevelopment and cognition among children and young adults with DS (Hart et al., 2017). Antioxidants, memantine, nicotine, and vaccination

against A-beta protein have also been trialed to target dementia in DS, but none have resulted in prevention of further decline or improvement in functioning (Hart et al., 2017).

There are several potential factors contributing to the lack of efficacy seen in clinical trials of pharmacological agents in DS to date. Given that neurodevelopment begins prenatally, interventions targeting older children and adults may miss critical windows of potential impact, and future studies directed toward even younger populations (infants, fetuses) may hold greater promise for influencing cognition (Stagni, Giacomini, Guidi, Ciani, & Bartesaghi, 2015). Better identification and analysis of subgroups is needed given the high level of inter-individual variability of cognitive and adaptive functioning in DS (Karmiloff-Smith et al., 2016; Liogier d'Ardhuy et al., 2015; Thomas et al., 2020).

Challenges capturing meaningful change with the use of current standardized outcomes measures have also impacted prior trials. Fortunately, this has led to large, coordinated efforts to identify more meaningful and valid measures that address issues with practice, ceiling, and floor effects, and which are suitable for use among a variety of specific age ranges and abilities (Esbensen et al., 2017; Liogier d'Ardhuy et al., 2015). Additionally, appropriate modification of materials and administration practices, rater training, and use of familiar tools and technologies can help to minimize anxiety and behavioral challenges that may contribute to underestimation of cognitive abilities in this population (Hart et al., 2017). Finally, early engagement with families in the research process may help in the formulation of better criteria for what constitutes a meaningful treatment benefit among individuals with DS (Heller et al., 2006). Continued collaboration between individuals with DS, families, medical professionals, researchers, industry partners, and advocacy organizations will be critical to move the field forward in the development of effective treatments to improve overall functioning among individuals with DS.



4. Developmental, cognitive, and learning profile

Collectively, individuals with DS have undergone extensive neuro-behavioral and cognitive phenotypic characterization over the last half-century (Abbeduto, Warren, & Conners, 2007; Carr, 1970; Carr & Collins, 2018; Edgin, 2013; Fidler, 2006; Fidler, Hepburn, & Rogers, 2006; Fidler & Nadel, 2007; Silverman, 2007). However, researchers and clinicians have consistently noted substantial inter-individual variability seen

in DS, with a wide range of abilities across developmental domains. A clear pathophysiologic understanding of what drives individual variation in DS remains elusive.

A prominent and nearly universal feature in DS is intellectual disability (ID), with cognitive abilities ranging from borderline abilities to profound cognitive impairment, with most individuals falling in the mild to moderate range of ID (Contestabile, Benfenati, & Gasparini, 2010; Vicari, Bellucci, & Carlesimo, 2005). IQ testing typically falls between 30 and 70, though cognitive abilities often do not develop evenly in DS. In general, the profile reveals relative strengths in visual processing and nonverbal cognitive abilities compared to verbal cognitive abilities (Contestabile et al., 2010; Lott & Dierssen, 2010; Næss, Lyster, Hulme, & Melby-Lervåg, 2011; Nelson et al., 2005). There are many limitations to standardized cognitive assessments including floor effects, motivation, compliance, and attention effects, leading to concerns that the most commonly used tools do not capture meaningful brain-based functional differences that may represent more appropriate outcomes measures for therapeutic trials (Edgin et al., 2010; Liogier d'Ardhuy et al., 2015). This has prompted research into tools designed specifically to assess outcomes in this unique population (Basten et al., 2018; Edgin et al., 2010).

In general, individuals with DS have a unique cognitive trajectory and continue to make cognitive gains through childhood and into early adulthood. Nonverbal skill acquisition is typically more consistent than verbal skill acquisition, which tends to decelerate in adolescence (Næss et al., 2011). However, overall cognitive growth should continue throughout childhood and adolescence, and any concern for loss of skills should warrant further evaluation and treatment.

4.1 Language

In infancy, delays are seen with acquisition of many early language skills, including canonical babbling, nonverbal requesting behaviors, acquisition of first words, combinations of multi word phrases and even gesture-word combinations (Buckley & Oliver, 1994; Iverson, Longobardi, & Caselli, 2003; Levy & Eilam, 2013). As they get older, children display expressive language abilities that fall far behind their receptive speech abilities, but use of early signing and a total communication approach may expand ultimate spoken vocabulary and improve functional communication abilities (de Barbosa et al., 2018; Özçalışkan, Adamson, Dimitrova, Bailey, & Schmuck, 2016).

As children approach preschool ages, they may struggle with articulation and intelligibility of spoken word, phonological errors, decreased growth of vocabulary, and growth of mean length of utterance (Chapman, 2006; Levy & Eilam, 2013). Single word vocabulary and pragmatic language are general strengths, but difficulties with syntax and phonological processing can encumber use of spoken word, auditory comprehension, and progression with reading skills which persists throughout childhood and adulthood (Chapman, 2006; Chapman & Hesketh, 2001; Couzens, Cuskelly, & Haynes, 2011; Levy & Eilam, 2013). There is a relative strength in single word receptive vocabulary that continues to grow over time, along with semantic understanding and conversational style, both of which tend to fall at or above the range of expected for overall mental age (Estigarribia, Martin, & Roberts, 2012; Levy & Eilam, 2013; Pulsifer, 1996). However, difficulties with syntactic understanding persist and magnify into adolescence. Auditory language comprehension is generally weak relative to typically developing peers, and individuals often benefit from heavy use of contextual cues to help them fully understand what is being said (Comblain & Rondal, 1996). Additionally, individuals with DS generally have poor monitoring of verbal comprehension abilities and are less likely to request clarification of instruction (Abbeduto et al., 2008). For these reasons, use of visual supports and a total communication approach are recommended.

4.2 Learning

Individuals with DS have a distinctive learning profile which has implications for therapeutic and educational programming. Rate and range of learning in DS may be highly variable with a trajectory that is discrepant from those who are typically developing and from those who have ID of other etiologies, related to characteristically altered patterns of short-term/working memory and long-term memory (Godfrey & Lee, 2018; Grieco, Pulsifer, Seligsohn, Skotko, & Schwartz, 2015; Vicari et al., 2005). Working memory describes the ability to hold, sort, process and manipulate information in the mind. Individuals with DS tend to have stronger visual-spatial working memory with more impairments in auditory working memory (Abbeduto et al., 2008; Chapman, 2006; Costanzo et al., 2013; Fidler, 2006; Fidler & Nadel, 2007; Lanfranchi, Jerman, & Vianello, 2009; Levy & Eilam, 2013). Visual processing, visuospatial short-term memory, and visual construction abilities appear to be commensurate with overall mental age (Couzens et al., 2011; Kogan et al., 2009).

Additionally, individuals with DS tend to excel with implicit learning and memory with procedural tasks such as watching the actions of another

and then imitating a skill being taught (Reed, Staytom, Stott, & Truzoli, 2011; Vicari, Verucci, & Carlesimo, 2007). However, they may struggle with explicit or declarative memory and tasks such as learning facts, knowledge, and concepts. There seem to be difficulties with encoding, storage, and later retrieval of these memory processes, which may be further negatively impacted by attention difficulties and higher processing demands (Kogan et al., 2009; Lanfranchi, Cornoldi, & Vianello, 2004). Explicit memory is critical to academic learning, and challenges in this area undoubtedly contribute toward difficulty with progress in academics.

Consolidation of memories may also be deficient considering known temporal lobe and hippocampal dysfunction (Lott & Dierssen, 2010) and likely extends beyond mere deficits in language. Additionally, learning tends to be maximized with socially oriented learning approaches and use of visual learning strategies. For these reasons, use of social stories, video-modeling, and hands-on learning strategies tend to be very effective teaching approaches.

4.3 Attention/executive function

Executive functioning impairments have also been described in DS. Impairments in visual selective attention and auditory sustained attention beyond those observed in mental-age matched TD controls and individuals with ID of different etiology have been described (Costanzo et al., 2013). There is poor verbal-response inhibition and inhibition of irrelevant information throughout childhood and into adulthood (Gilmore & Cuskelly, 2009; Lott & Dierssen, 2010). Executing actions in planning tasks takes longer and shifting attention between tasks (especially verbal tasks) can lead to inefficient adaptation to different situations (Landry, Russo, Dawkins, Zelazo, & Burack, 2012; Lanfranchi, Jerman, Dal Pont, Alberti, & Vianello, 2010). There are also difficulties with simultaneous processing of information (Lanfranchi et al., 2009).

4.4 Social-emotional characteristics

In children with DS, social development ordinarily occurs in a similar fashion and sequence compared to that of typically developing children, though at a slower pace. Children with DS tend to seek out social interaction more frequently than individuals with ID of other etiologies (Jahromi, Gulsrud, & Kasari, 2008). Relative to mental-age matched typically developing peers, young children with DS display strengths in nonverbal social interaction skills such as joint attention, declarative pointing, and eye contact, but tend to be weaker with nonverbal requesting behaviors for instrumental

purposes, such as to request for objects or assistance with objects, which also tends to correlate with overall language ability and future cognition (Fidler, Philofsky, Hepburn, & Rogers, 2005; Mundy, Sigman, Kasari, & Yirmiya, 1988).

Children with DS also have strengths in imitation abilities, pro-social empathetic behaviors, social motivation, and social engagement, though challenges may come with social problem-solving and higher-order social cognition tasks (Fidler, 2006; Fidler & Nadel, 2007; Kasari, Freeman, & Bass, 2003). Children with DS with poorer early infant regulatory behavior may be at higher risk for social communication challenges (Schworer, Fidler, & Daunhauer, 2021). Joint attention, emotional recognition, social referencing, spontaneous gesturing, requesting for social purposes, and other social communication skills should be monitored closely through early childhood as these areas are notably atypical among children with DS and co-occurring autism spectrum disorder (ASD).



5. Hypotonia and motor impairment

Hypotonia refers to decreased resistance to passive muscle stretch or movement and contributes to hyperextension at the joints. Tone is determined subjectively by physical examination and is difficult to evaluate independently from joint hyperlaxity and hypermobility. Both ultimately play a part in gross motor delays seen in children with DS (Foley & Killeen, 2019). Individuals with DS often display over-pronation and flat feet (pes planus), which play a role in delayed walking and may cause potential long-term alignment and joint issues leading to easy fatiguing, instability of gait, and the possible development of pain throughout life (Agiovlasitis, McCubbin, Yun, Pavol, & Widrick, 2009; Foley & Killeen, 2019). Low muscle tone can also contribute to other clinical symptoms such as generalized difficulties with postural control, breathing difficulties, feeding problems, excessive drooling, and articulation difficulties.

Motor development in DS proceeds sequentially in the typical fashion, but along a delayed timeline and with more inter-individual variability. Longitudinal studies of children with DS reveal that they require more time to achieve walking (about 50% walk by age 24 months) and running (about 50% run by 5 years of age), with more falling or difficulty navigating obstacles or uneven surfaces (Winders, Wolter-Warmerdam, & Hickey, 2019). Individuals with DS tend to display gait instability, clumsiness, poor refinement of movements, imbalance, generalized weakness, low stamina, slower reaction times, and slower times to completion of a

movement (Luz Carvalho & Adami Vasconcelos, 2011). Hypotonia and subsequent gait instability contributes to lower overall levels of physical activity which can impact bone density, body mass index, and other aspects of general health.

Some research shows that motor development may influence cognitive and social-emotional development (Leonard, 2016). For example, earlier ambulation is associated with improved concepts of object permanence, social referencing, pointing and gaze shifting, and spatial cognition (Libertus & Hauf, 2017; Virji-Babul, Kerns, Zhou, Kapur, & Shiffar, 2006). One theory is that inefficient locomotion may limit the way a child can explore their surroundings, requiring increased attention and energy expenditure on the process of moving about itself which may draw attention away from stimulating objects in the environment (Iarocci, Virji-Babul, & Reebye, 2006; Virji-Babul et al., 2006). Clinically, parents of infants with DS often report that walking is one of their most valued goals, and research has demonstrated increases in independence and positive affect during play in more mobile children, and lower stress levels among children who were more self-sufficient with mobility (Biringen, Emde, Campos, & Appelbaum, 1995; Sloper, Knussen, Turner, & Cunningham, 1991; Ulrich, Ulrich, Angulo-Kinzler, & Yun, 2001). Additionally, stable and efficient locomotor skills are of social relevance, as mobility is paramount to participation in games and activities with other children.

Continued strengthening and stability can help to offset some of the consequences of motor challenges. Physical therapy is one of the mainstays of early intervention and can help to accelerate muscle strengthening, enhance stability and balance, and improve motor planning (Ruiz-González, Lucena-Antón, Salazar, Martín-Valero, & Moral-Munoz, 2019). Treadmill training with alternating stepping practice has been shown to improve acquisition of locomotor milestones, muscle strengthening, and postural control (Ulrich et al., 2001). Physical equipment such as shoe inserts and orthotics (e.g., supra malleolar orthotics—SMOs, ankle-foot orthotics—AFOs) help to support the foot and ankle and improve alignment. Children may benefit from physical therapy at school and in the community throughout childhood to improve functional motor abilities.

Fine motor skills are also delayed in DS, which has implications for independence in daily living skills (Beqaj, Tërshnjaku, Qorolli, & Zivkovic, 2018; Frank & Esbensen, 2015). For example, children with DS have been found to have decreased grip and pinch strength (John, Dhanve, & Mullerpatan, 2016). Occupational therapy can help children with DS to gain fine motor and adaptive skills through explicit teaching and repetition of

skills, provision of accommodations to support challenges with fine motor difficulties (e.g., adaptive clothing), and modifications of the environment.

Individuals with DS should continue to make progress with strengthening and stability throughout childhood into adulthood. Any change or loss of motor function should prompt investigation for musculoskeletal or neurological conditions that individuals with DS may be at higher risk of developing. Atlantoaxial instability may present with gait changes, new-onset torticollis or head tilt, neck pain, and decreasing mobility overall (Ferguson, Putney, & Allen, 1997; Pueschel & Scola, 1987). Symptoms of spinal cord impingement or abnormality should be screened for throughout childhood with any concerns warranting further evaluation via cervical x-ray and spinal MRI (Bull, 2011).

Neurological conditions may impact motor function. An important example is Moyamoya, which is a cerebrovascular abnormality in which the carotid artery becomes narrowed or blocked, reducing blood flow to the brain and causing many small abnormal blood vessels to develop. Moyamoya is more common in DS, and may lead to ischemic or hemorrhagic strokes or transient ischemic attacks which can present with seizures or sudden or intermittent focal neurologic deficits such as new onset weakness or paralysis and necessitates immediate neurological and neurosurgical evaluation (See et al., 2015).

Musculoskeletal or orthopedic causes should also be considered with any functional motor decline. Conditions such as arthropathy of DS (an inflammatory arthritis), subluxation of the hips, scoliosis, slipped capital femoral epiphysis, patellofemoral instability, or worsening pes planus or other foot disorders can occur in DS (Caird, Wills, & Dormans, 2006; Foley & Killeen, 2019; Foley, Deely, MacDermott, & Killeen, 2019). Involvement of orthopedics, neurosurgery, rheumatology, physiatry, physical therapy, and orthotic services can be essential for appropriate management of these conditions that impact motor function in DS.



6. Impact of medical conditions on neurodevelopment

DS can affect nearly all body systems and is associated with an increased risk of many non-neurologic medical conditions that may impact neurodevelopment (Bull, 2011). Potential co-occurring conditions include congenital heart disease (CHD), iron deficiency anemia, hypothyroidism, celiac disease, and hearing and vision deficits. These are conditions that should be avidly screened for, promptly identified, and comprehensively

managed to minimize negative impacts on neurodevelopment. Additionally, sleep dysfunction and obstructive sleep apnea are prevalent and impactful enough in DS that a separate section is dedicated to their discussion.

6.1 Neonatal conditions

Prematurity and lower birth weight are both more common in DS, and as in the general population, these factors may adversely impact neurodevelopment (Brock, Walsh, & Allen, 2021; Pascal et al., 2018). For example, in the general population, prematurity and low birthweight are associated with increased risks of executive functioning difficulties, ASD, cognitive impairment, and cerebral palsy (Crump, Sundquist, & Sundquist, 2021; Pascal et al., 2018). Some congenital abnormalities common in DS may require early surgical intervention and prolonged hospitalization which may impact neurodevelopment (Stoll, Dott, Alembik, & Roth, 2015). Some of these conditions include certain types of CHD (e.g., complete atrioventricular canal defects) and gastrointestinal malformations (e.g., duodenal atresia, Hirschsprung's disease, trachea-esophageal fistulas). Though research on developmental outcomes among children with DS and these risk factors is not well-developed, some research has not shown a significant impact on cognitive outcomes (Rosser et al., 2018).

6.2 Congenital heart disease

In the general population, CHD is associated with impacts on executive functioning, visuospatial skills, memory, mathematics abilities, and language development by school-age (Bellinger et al., 2011; Rollins & Newburger, 2014). Formal neurodevelopmental diagnoses of attention deficit hyperactivity disorder (ADHD) and learning disabilities occur with increased frequency in those with CHD (Bellinger et al., 2011; Rollins & Newburger, 2014). In general, milder forms of CHD have fewer neurodevelopmental sequelae, while more complex lesions may cause more severe neurological deficits (Marino et al., 2012; Rollins & Newburger, 2014). In DS specifically, some studies have examined neurodevelopmental outcomes in children undergoing cardiac surgery during the first year of life and have not revealed any long-term differences at school age and beyond compared to children with DS without CHD (Alsaied et al., 2016; Rosser et al., 2018; Visootsak et al., 2011, 2016). Further research is needed, including investigation of outcomes for children with DS with non-operative CHD.

6.3 Hypothyroidism

Because thyroid hormone is essential for brain development, children with untreated congenital or early hypothyroidism may be at risk for impaired brain development and subsequent cognitive and motor deficits (Bernal, Guadaño-Ferraz, & Morte, 2003; Klein, Meltzer, & Kenny, 1972; Salazar, Cisternas, Martinez, & Inestrosa, 2019). Thyroid hormone is critical for development and differentiation of cells in the nervous system, and in DS, untreated hypothyroidism may have a deleterious effect on an already at-risk developing nervous system.

Thyroid abnormalities are typically quickly controlled with the use of thyroid hormone supplementation. Subclinical hypothyroidism, with elevated TSH values but normal T4 levels, is also commonly seen in DS and is often transitory when diagnosed in the first 3 years of life (Claret et al., 2013). Whether this phenomenon should be monitored closely for development of overt hypothyroidism vs. treated with thyroid hormone supplementation is unclear, though older age and presence of thyroid antibodies or a goiter may signify higher risk of future overt hypothyroidism (AlAaraj, Soliman, Itani, Khalil, & De Sanctis, 2019; Claret et al., 2013). In one study, a cohort of 24-month-old children with DS and subclinical hypothyroidism were treated with thyroxine hormone supplementation, and an initial small improvement in motor development, length, and weight were seen relative to an untreated control group (van Trotsenburg et al., 2005). However, when these children were examined in mid-childhood, there were no significant differences in any measures of development (Marchal et al., 2014). It should be noted that there is no research that reveals negative developmental consequences of treated hypothyroidism.

6.4 Iron deficiency anemia (IDA)

Iron deficiency Anemia has been associated with motor and cognitive developmental deficits in the general population and may be irreversible, leading to changes in neurotransmitter homeostasis, decreased myelin production, impaired synaptogenesis, and declining function of the basal ganglia (Agaoglu, Torun, Unuvar, Sefil, & Demir, 2007; Pivina, Semenova, Doşa, Dauletyarova, & Björklund, 2019; Tenenbaum et al., 2011). Additionally, in the general population, anemia is a frequent comorbidity in ADHD and ASD (McCann & Ames, 2007; Pivina et al., 2019). Children with DS are at risk for iron deficiency anemia at rates similar to or higher than the general population, although there are no studies looking

at cognitive outcomes specifically among children with DS and iron deficiency. Of note, iron deficiency may be harder to detect in DS due to the common finding of larger sized red blood cells in DS (macrocytosis). Therefore, annual screening for anemia, iron insufficiency, and iron deficiency with more specific iron studies is recommended in DS (Bull, Trotter, Santoro, Christensen, & Grout, 2022; Hart et al., 2020), and when detected, iron should be replenished following similar protocols to those used in the general population.

6.5 Celiac disease

Celiac disease, which is also more common in DS, is an autoimmune disorder affecting the gastrointestinal system. Celiac disease results from an immunologic reaction to gliadin and related proteins of wheat, rye, and barley that leads to inflammation of the small intestine. Compared to the general population, the symptomatic form of celiac disease appears to be more frequent in children with DS than the asymptomatic form. However, some research has shown that in adolescents or adults with DS and celiac disease, one-third may have no gastrointestinal symptoms, with 20% having silent disease or 10% showing extra-intestinal manifestations, including dermatologic manifestations, neurobehavioral symptoms, overt neurologic symptoms like cerebellar ataxia, peripheral neuropathy, or seizures (Bonamico et al., 2001; Jericho & Guandalini, 2018; Pavlovic, Berenji, & Bukurov, 2017).

Prompt identification and treatment of celiac disease are important for improving outcomes in DS. Children with DS and celiac disease often present with the more classic gastrointestinal symptoms including diarrhea, vomiting, abdominal distention, pain, fatigue, poor weight gain, and behavior changes. Because these symptoms are very common in DS, and because the evaluation of behavior changes among individuals with ID may be challenging, diagnostic overshadowing can lead to missed diagnoses of celiac disease. Late identification or missed diagnosis of CD can have important health and neurodevelopmental implications, such as failure to thrive, anemia, osteoporosis, lymphoma, and neurologic symptoms (Hadjivassiliou, Duker, & Sanders, 2014; Jericho & Guandalini, 2018). It is important to assess for symptoms of celiac disease regularly in individuals with DS, and, if present, to proceed with further lab-based screening evaluation typically with tissue transglutaminase immunoglobulin A (tTG-IgA) and total IgA (Bull et al., 2022). If screening is positive, the diagnosis of celiac disease

can be confirmed with duodenal biopsy. Treatment includes institution of a strict gluten-free diet, which can be accomplished with assistance from nutritional counseling and behavioral support. In the general population, gluten-free dietary treatment leads to resolution of small intestine histologic inflammation and improvement in gastrointestinal symptoms as well as neurologic symptoms such as ataxia and ADHD symptoms, though further research on improvements in “soft neurologic” symptoms is underway and should also be more closely examined in DS (Mearns et al., 2019; Niederhofer, 2011).

6.6 Hearing and vision impairments

Hearing loss is common in DS, and even minor hearing loss may negatively impact speech and language development, especially in the setting of known phonological and speech sound difficulties and general auditory processing deficits in DS (Laws & Hall, 2014). This is especially important between birth and 5 years of age when language acquisition is occurring at its most rapid pace. Middle ear disorders are very common among young children with DS due to anatomic differences which can lead to recurrent infections and effusions and subsequent conductive hearing loss (Bull, 2011; Kreicher, Weir, Nguyen, & Meyer, 2018). Additionally, there can be long-term chronic effects of middle ear effusions with worsened speech discrimination and performance on language comprehension tasks well into the adolescent and adult years, necessitating appropriate early management (Deggouj et al., 2012; Okada, Welling, Liberman, & Maison, 2020). Risk of sensorineural hearing loss is lower than risk of conductive hearing loss in children, but still much higher than in the general population (De Schrijver, Topsakal, Wojciechowski, Van de Heyning, & Boudewyns, 2019). As such, frequent hearing testing during early childhood is necessary in coordination with otolaryngology when the tympanic membrane cannot be clearly visualized (Bull, 2011). Hearing screening should be continued on a yearly basis into adulthood, when early onset of age-related hearing loss (presbycusis) can also occur (Buchanan, 1990).

Vision abnormalities can similarly impact developmental trajectories, skill acquisition, school participation, and behavior when not addressed appropriately. Individuals with DS are at increased risk for many treatable ophthalmologic conditions such as refractive errors, strabismus, and cataracts (Postolache, Monier, & Lhoir, 2021). Adequate visual functioning is essential for learning, and early assessment and management can enhance quality

of life. The AAP recommends ophthalmology evaluations for children with DS on a regular basis (Bull, 2011). Glasses are often needed. Special frames and bands may be helpful for fit given facial structural differences, and behavioral strategies may promote adherence to glasses wearing. Vision problems should also be considered anytime there is a change in behavior or problems in school.



7. Sleep dysfunction

Compared to the general population, sleep problems are reported more frequently in DS, occurring among 31–54% of children (Stores & Stores, 2013; Tietze et al., 2012). In DS, abnormalities of sleep include remarkably high rates of obstructive sleep apnea, frequent insomnia (both with sleep onset and sleep maintenance), longer overall time in bed, decreased sleep efficiency and time in the rapid eye movement (REM) phase of sleep, increased movement, more fractured sleep patterns, and increased rates of abnormal sleep arousals (Ashworth, Hill, Karmiloff-Smith, & Dimitriou, 2013, 2015; Harvey & Kennedy, 2002). Circadian rhythms and phases do appear to be relatively preserved among children with DS and, similar to the general population, increased sleep efficiency and consolidation emerge with increasing age (Fernandez et al., 2017).

Because up to 50–80% of children with DS exhibit evidence of obstructive sleep apnea, any assessment of sleep dysfunction in DS typically starts with evaluation for sleep disordered breathing, especially given its potential impact on neurodevelopment (Maris, Verhulst, Wojciechowski, Van de Heyning, & Boudewyns, 2016; Nerfeldt & Sundelin, 2020; Shott et al., 2006). Obstructive sleep apnea classically presents with heavy breathing, snoring, apneic pauses, frequent night awakenings, unusual sleep positions, daytime sleepiness, and behavior problems. Many young children with DS are asymptomatic by parental report but are found to have obstructive sleep apnea on sleep study (Nerfeldt & Sundelin, 2020; Shott et al., 2006). Craniofacial differences (midface and maxillary hypoplasia), relative macroglossia, upper airway abnormalities, relative hypotonia with increased rates of laryngomalacia, and higher rates of obesity during adolescence are all factors that contribute to this higher prevalence of obstructive sleep apnea (Chamseddin, Johnson, & Mitchell, 2019).

Several studies have shown similar, if not more striking, neurodevelopmental ramifications of obstructive sleep apnea among children with DS compared to the general population. Even mild obstructive sleep

apnea has been associated with significant decline in verbal IQ scores, worsened cognitive flexibility, poorer visual-perceptual skills, increased behavioral symptoms and disruptive behaviors, poorer overall developmental milestone acquisition, and worsened school performance (Chen, Spanò, & Edgin, 2013; Churchill, Kieckhefer, Bjornson, & Herting, 2015; Joyce & Dimitriou, 2017; Lee et al., 2020; Nixon, Biggs, Jitpiriyaroj, & Horne, 2016; Stores & Stores, 2013). With increasing severity of obstructive sleep apnea, the likelihood of behavioral abnormalities among children with DS increases, with higher rates of inattentive, impulsive, and hyperactive behaviors and lower overall developmental assessment scores (Anand et al., 2021).

Health care providers should engage in frequent screening for symptoms of obstructive sleep apnea among all children and adolescents with DS (Bull, 2011). In light of the clear impact of sleep-disordered breathing on development and behavior (as well as cardiopulmonary health) and the lack of correlation between parent-reported symptoms and presence of obstructive sleep apnea, the AAP recommends that all children with DS undergo a sleep study by age 4 years regardless of symptoms (Bull, 2011). Follow-up sleep studies may be warranted at other points in childhood, adolescence, or adulthood depending on symptoms, weight gain, and other co-occurring medical conditions.

There are multiple options for treatment of obstructive sleep apnea. Adenotonsillectomy is usually the first-line treatment for children with DS, as medications for mild symptoms (nasal steroids, montelukast) in DS may be less effective than in the general population (Yu et al., 2020). However, even after adenotonsillectomy, there may be residual obstructive sleep apnea such that repeat sleep study after surgical intervention is necessary (Nation & Brigger, 2017; Nerfeldt & Sundelin, 2020). Continuous Positive Airway Pressure (CPAP) is typically recommended for residual obstructive sleep apnea, and behavioral intervention and desensitization approaches can be beneficial for those who have difficulty tolerating CPAP. When these approaches are not sufficient, or when CPAP cannot be tolerated, further investigation via drug-induced sleep endoscopy and higher-level upper airway surgery may be indicated. Other treatment options may include dental surgical approaches and other emerging therapeutic possibilities including hypoglossal nerve stimulation (Simpson, Oyekan, Ehsan, & Ingram, 2018). Finally, generalized weight loss is generally recommended as there is an association between obesity in DS and the development of obstructive sleep apnea, especially in adolescents

and young adults (Bertapelli, Pitetti, Agiovlasitis, & Guerra-Junior, 2016; Manuel, Shah, Mitchell, & Johnson, 2021).

Beyond sleep disordered breathing, several other organic sleep disorders may be elicited through symptomatic screening, questionnaire data, or collection of data via sleep study or actigraphy. Arousals that accompany sleep (sleepwalking, sleep terrors, excessive nightmares) tend to be more commonly endorsed by parents of children with DS (Ashworth et al., 2013; Breslin, Edgin, Bootzin, Goodwin, & Nadel, 2011; Stores & Stores, 2013). Most arousals are benign in nature and do not necessarily warrant treatment but may be very disconcerting or disruptive to family members, meriting thorough anticipatory guidance and education on sleep hygiene. Sleep-related movement disorders include rhythmic movements of the legs, limbs, or jaw during sleep such as periodic limb movement disorder, restless leg syndrome, or teeth grinding, all of which are more common in DS (Rosen, Berbert, & Weller, 2020). There is an association of restless sleep with iron deficiency, so management typically starts with obtaining a ferritin level and beginning iron supplementation for ferritin levels <50ng/dL (Leung, Singh, McWilliams, Stockler, & Ipsiroglu, 2020). If symptoms are very severe, use of dopamine agonists and calcium channel alpha-2 antagonists are first line pharmacologic treatments, though these have not been studied specifically in DS (Silber et al., 2021).

Behavioral sleep challenges generally involve insomnia that is due either to sleep onset difficulties or to sleep maintenance difficulties or a combination of both. Parents of children with DS note significant difficulties with settling and falling asleep, anxiety and resistance surrounding bedtime, and challenges with sleep maintenance through the night with increased night-time awakenings (Ashworth et al., 2013; Breslin et al., 2011; Cotton & Richdale, 2006; Esbensen & Schwichtenberg, 2016). Insomnia in DS is associated with poorer executive functioning and verbal fluency while daytime sleepiness may lead to poor inhibitory control and behavioral issues with increased irritability and engagement in stereotypies (Chen et al., 2013). Children with DS may also seek out more parental presence at sleep-onset and more room-sharing which are associated with less cumulative night-time sleep (Yau et al., 2019). This sleep dysfunction increases parental stress and impacts family functioning, suggesting that programs targeting self-soothing and sound sleep-hygiene practices from an early age are an important aspect of treatment (Churchill et al., 2015; Esbensen, Schworer, Hoffman, & Wiley, 2021; Fernandez et al., 2017; Yau et al., 2019).

A thorough history of sleep problems, timing, parental responses, and psychosocial triggers along with the use of sleep diaries and standardized questionnaires such as the Childhood Sleep Habits Questionnaire (CSHQ) can clarify the scope of sleep-related dysfunction (Esbensen & Hoffman, 2017; Hoffmire, Magyar, Connolly, Fernandez, & van Wijngaarden, 2014). Education regarding sleep hygiene is critical, such as reducing night-time screen time, establishing consistent bedtime routines, and maintaining consistent sleep wake-schedules. Behavioral strategies may be very effective, such as planned ignoring of bedtime resistance and attention-seeking behaviors, fading adult support with sleep onset, strategic napping, and use of positive reinforcement systems such as the bedtime pass (Esbensen, Beebe, Byars, & Hoffman, 2016; Friman et al., 1999; Moore, Friman, Fruzzetti, & MacAleese, 2007), though few behavioral insomnia treatments have been systematically studied in DS.

Use of pharmacologic interventions generally has variable effectiveness among children with behavioral sleep dysfunction, especially among those with neurodevelopmental disabilities (Bruni, Angriman, Melegari, & Ferri, 2019). Medications are typically more effective in addressing sleep onset, while sleep maintenance is more difficult to treat pharmacologically. Melatonin is often a first-line option for sleep-onset related insomnia in childhood, including in those with DS and other neurodevelopmental disabilities (Bruni et al., 2015, 2019; Jan, Espezel, & Appleton, 1994). Other options may include antihistamines (e.g., diphenhydramine, hydroxyzine), alpha-adrenergic agonists (e.g., clonidine, guanfacine), antidepressants (e.g., mirtazapine, trazodone) and other medications that have sedating properties (e.g., gabapentin, quetiapine) (Bruni et al., 2019). All of these medications should be used with caution and require close monitoring given limited published safety data, though clinically many of these medications may be at least partially effective and are usually well-tolerated. Benzodiazepine use for sleep dysfunction in children has limited utility and generally should not be considered.



8. Epilepsy

Children with DS have an increased risk of epilepsy, with prevalence rates ranging from about 1–13% (Arya, Kabra, & Gulati, 2011). Epilepsy in DS is age specific, with a first peak in incidence in infancy or early childhood (most commonly Infantile (Epileptic) Spasms (IS)), another peak in early adulthood, and a final peak typically in the 5th decade of life or later

("Late Onset Myoclonic Epilepsy in DS (LOMEDS)"), which is associated with Alzheimer's disease (Arya et al., 2011). This review will focus on epilepsy of childhood. All major seizure types have been described in DS (Altuna, Giménez, & Fortea, 2021; Arya et al., 2011; Goldberg-Stern et al., 2001; Stafstrom & Konkol, 1994). This includes IS, febrile seizures, generalized seizures, and partial seizures. Febrile seizures may occur less often in DS than in the general population, whereas unprovoked seizures are more common in children and young adults with DS (Altuna et al., 2021). The reason for increased risk of epilepsy in DS is largely unknown, and may be related to inherent genetic features, structural brain differences, altered synaptogenesis, membrane channel dysfunction, or acquired causes such as stroke, hemorrhage, or hypoxic or ischemic injury secondary to complications from other health conditions like moyamoya syndrome, congenital heart disease, or infection (Arya et al., 2011; Goldberg-Stern et al., 2001; Stafstrom & Konkol, 1994). The diagnosis, classification, and treatment approaches for epilepsy in children and young adults with DS does not differ from the general population, though many antiepileptic medications have not been specifically studied in DS. It should be noted that baseline abnormalities on EEG, even in the absence of seizures, have been described in DS (Arya et al., 2011; Śmigielska-Kuzia, Sobaniec, Kuśak, & Boćkowski, 2009), but this does not necessarily warrant treatment in the absence of clinically concerning events.

IS are the most common epilepsy syndrome in children with DS, occurring in 0.6–13% (and accounting for about 40% of epilepsy in DS) (Arya et al., 2011). IS typically occur in the first year of life, on average between 6 and 8 months of age. However, some children may present as late as 18 months of age (Arya et al., 2011; Harvey et al., 2022), and presentations within the later end of this range may be seen more often in DS than in the general population (Marandi, Kumar, Kaddurah, & Chugani, 2017). IS are characterized by sudden abnormal movements of the body, usually involving unprovoked symmetric, synchronous flexion or extension of the trunk and stiffening of the arms and legs, that can occur in clusters, often during sleep-wake transitions. IS can be subtle in infants with DS.

IS may be difficult to recognize in DS, and may be mistakenly attributed instead to gastroesophageal reflux, hypotonia, pain, myoclonus, shuddering, or other benign repetitive mannerisms in infants, all of which can also be evident in DS. For example, Sandifer syndrome is characterized by abnormal movements of the trunk associated with gastroesophageal reflux in early childhood. It can involve tonic posturing of the head, neck and trunk, or

writhing but non-rhythmic movements. The abnormal movements are typically associated with gastrointestinal symptoms and resolve with treatment of the reflux. Other paroxysmal movements which can be seen commonly in neonates and infants must also be considered in the differential diagnosis of IS. These include jitteriness (recurrent tremors in neonates, characteristically triggered by stimuli and suppressible), shuddering attacks (rapid shivering of the head, shoulders, and sometimes trunk), paroxysmal torticollis of infancy, breath-holding spells (which can be associated with body movements and posturing), benign neonatal sleep myoclonus or benign myoclonus of early infancy, self-stimulatory/gratification behaviors in infancy and childhood, or paroxysmal tonic upgaze (sustained or constant conjugate upward eye gaze sometimes associated with additional body movements). These are usually benign movement disorders that typically self-resolve but tend to prompt concerns among parents when observed and given the higher risk of epilepsy in DS, should be evaluated with EEG or other diagnostic assessments.

IS are a form of epileptic encephalopathy, usually associated with significant neurodevelopmental consequences such as change in behavior, developmental plateau or arrest, or frank regression or loss of skills involving motor, communication, and/or social skills (Shields, 2006). Some research has shown developmental regression present in more than half of those with DS and IS (Harvey et al., 2022). The developmental concerns may be the symptom that prompts families to seek initial evaluation rather than the spasms themselves. Developmental concerns may include decreased babbling, less social responsiveness (eye contact, smiling, response to name), deterioration of head control and tone, or increased fussiness/irritability. Sometimes, these symptoms are misattributed to expected developmental delays in DS, or to other causes such as hearing or vision problems.

The diagnosis of IS is made by electroencephalogram (EEG), which shows a characteristic pattern of abnormal brain waves arising from multiple regions throughout the brain. The diffusely disorganized, chaotic background activity on EEG, called hypsarrhythmia or modified hypsarrhythmia, confirms the diagnosis. This is typically more pronounced in sleep, necessitating an EEG study that captures sleep, especially early in the course of IS. When spasms are captured on EEG they often result in a distinctive pattern change in the brain waves as well. Although brain MRI is typically part of the standard evaluation of IS in those without DS, minimal neuroimaging data exists in DS and IS. Structural etiology was rarely identified in one study of DS and IS (Trowbridge et al., 2019).

Various treatment regimens have been used to treat IS, but hormonal treatment with ACTH or steroids is typically the preferred first-line treatment with high response rates described in DS (Sanmaneechai et al., 2013). Vigabatrin or other antiepileptic treatments have also been used, and recently there has been some investigation of combination treatment of Vigabatrin with hormonal treatment (Harvey et al., 2022; O'Callaghan et al., 2017, 2018), although there is no definitive evidence that this is favorable in DS. Treatment should be started immediately, with a goal of seizure cessation and resolution of hypsarrhythmia as quickly as possible. Frequent EEG and close clinical monitoring are needed to gauge response to treatment. When children do not respond to first-line treatments, other treatment options such as the ketogenic diet or additional antiepileptic medications should be considered.

Some studies have shown more favorable seizure outcomes and a lower risk of developing subsequent epilepsy in DS compared to those who have IS without DS (Beatty, Wrede, & Blume, 2017; Goldberg-Stern et al., 2001). However, other studies have shown persistent seizures (Harvey et al., 2022) and the potential for a late relapse, up to 2 years after the onset of IS (Sanmaneechai et al., 2013). Thus, there is variability in seizure outcomes in DS. More research is needed to understand factors associated with IS outcomes. Seizures may elapse or evolve into Lennox-Gastaut syndrome, which is characterized by intractable epilepsy with several different seizure types and a characteristic EEG pattern that can occur even in the absence of preceding IS (Arya et al., 2011). In DS, Lennox-Gastaut syndrome can be associated with a predominance of reflex seizures which are precipitated by sudden sensory stimulation such as noise or touch (Arya et al., 2011).

Even with proper treatment and swift seizure control, IS can have a detrimental effect on development. IS have been associated with developmental delays, later cognitive impairment, and a higher risk of autism spectrum disorder (Sanmaneechai et al., 2013; Tapp, Anderson, & Visootsak, 2015). However, it is difficult to assess the neurodevelopmental impact of IS in children with DS because of pre-existing developmental delays and the wide range of cognitive, language, and social abilities seen in DS. Factors that have been suggested to result in worse neurodevelopmental outcomes in children with DS have included delayed recognition and initiation of treatment which may occur more commonly in DS, poor seizure outcomes (persistent or recurrent seizures), and the presence of significant developmental regression for a prolonged time prior to treatment initiation (Esbensen, Rojahn, Aman, & Ruedrich, 2003). However, not all research studies have shown

that these factors contribute to differences in developmental outcomes and such studies on developmental outcomes have typically included small subject numbers, indicating that further research is needed in this area.



9. Abnormal movements

Abnormal movements are diverse and common in DS (Dinan & Golden, 1990; Haw, Barnes, Clark, Crichton, & Kohen, 1996). Many such movements can be described as paroxysmal, occurring intermittently with return to baseline between episodes. There are several potential causes of paroxysmal movements, including those that are often transient developmental disorders, and those that are more persistent. Sleep-related movement disorders and epilepsy were discussed separately in previous sections.

9.1 Transient developmental movement disorders

Spasmus Nutans is an early childhood disorder associated with a triad of nystagmus, head nodding, and abnormal head position (torticollis or head tilt). The condition may be more common in DS (Wagner, Caputo, & Reynolds, 1990). It should be noted that isolated nystagmus is also quite common in DS (Postolache et al., 2021), as is torticollis due to fourth nerve palsy or non-ocular causes (Puig Galy, Galán Terraza, & Romera Becerro, 2006). In fact, abnormal head posture has been described in DS, and has many potential etiologies (Dumitrescu, Moga, Longmuir, Olson, & Drack, 2011). Retinal and intracranial conditions should be ruled out, though evidence of a progressive neurologic etiology for nystagmus has not been found in a review of neuroimaging in DS (Wong, Schneier, Ledoux, & Heidary, 2016). There are no specific treatments for spasmus nutans, which typically resolves spontaneously in early childhood.

9.2 Hyperkinetic movement disorders

Stereotypies are repetitive, non-functional movements that often follow a common pattern and may include whole body movements, such as rocking or twirling, or more isolated rhythmic movements of the head or limbs, such as finger wiggling, hand flapping, arm stiffening, or toe walking. The repertoire for each child can evolve over time, but the movements are often first seen during the infant or toddler years. Stereotypies tend to occur more when the child is stressed, excited, fatigued, bored, or when engrossed in an activity. The movements can be intermittent or can be continuous

and prolonged. They can be confused with complex motor tics but can be characteristically suppressed by distraction or initiation of another activity, and onset tends to be younger for stereotypies compared to tics. Stereotypies are common in neurodevelopmental disorders, including in those with ASD and ID (Singer, 2009).

In DS, stereotypies have been found to be common in those who do and do not have co-occurring ASD (Ji, Capone, & Kaufmann, 2011), however some research shows that the character and severity of stereotypies may differ in the presence of ASD (Carter, Capone, Gray, Cox, & Kaufmann, 2007). While the neurobiological basis for stereotypies is not known, one study in DS demonstrated a positive correlation between cerebellar white matter volumes and severity of stereotypic behaviors (Carter, Capone, & Kaufmann, 2008). There are limited evidence-based treatments for stereotypies. Stereotypies are thought to serve a sensory self-stimulatory function that is then automatically reinforced, and there is some evidence that behavioral approaches using differential reinforcement intervention may be effective at reducing these behaviors (Neil & Jones, 2015). Medications are generally not effective in the treatment of stereotypic movement disorders, though may be tried in certain cases, particularly if they are significantly impairing participation, social interaction, or general functioning. However, in most cases, stereotypies do not necessarily merit treatment.

Motor tics (repeated, non-rhythmic, intermittent movements) and vocal tics can also occur in DS and tend to be more brief and abrupt than stereotypies, less distractible, and usually have a later onset around 5–7 years of age. They may be precipitated by excitement or stress but tend to be more random and sudden when they occur. Tics are associated with a premonitory urge to do the movement, and often lead to inner tension and the desire to suppress the movements. However, these defining characteristics can be challenging to elicit from the medical history in DS. Vocal tics can also occur. Motor tics may be simple or complex and often involve the eyes, face, head, or shoulders such as blinking, grimacing, twisting, or shrugging. Tics may be transient or tend to wax and wane over time, and often improve or resolve by adulthood. Tics often co-occur with anxiety, obsessive-compulsive disorder (OCD), or ADHD. Tics can be treated with both behavioral and pharmacological approaches if impairing or cause distress to the child.

Other abnormal movements can also be seen in older children and adults. Oromotor dyskinesias, or involuntary repetitive movements of the mouth and face are particularly common in DS (Dinan & Golden, 1990; Haw et al., 1996). New onset of abnormal postures, stiffening, slow movements,

or abnormal limb movements, especially involving the neck and upper extremities should prompt consideration of atlanto-axial instability.



10. Challenging behaviors

Behavioral challenges are very common in children and adolescents with DS and may be diagnosed in approximately 1/3 of individuals with DS (Dykens, 2007). This may be lower compared to those with other cognitive disabilities (Chapman & Hesketh, 2000; Dykens, 2007). However, some research has shown that behavior concerns are reported by parents in up to 94% of children with DS (Patel, Wolter-Warmerdam, Leifer, & Hickey, 2018). Children with DS have been described as having a personality-motivation style, characterized by strong-willed or stubborn temperament (Fidler & Nadel, 2007; Fidler, 2006; Kasari & Freeman, 2001). Social motivation is often a strength, but children with DS may also block completion of requested tasks through social distraction (Kasari & Freeman, 2001), leading to noncompliant behaviors throughout childhood. Poor task persistence and decreased intrinsic motivation to interact with the environment in order to learn tends to be an individual characteristic noted early in childhood in DS, which often persists (Gilmore & Cuskelly, 2009). Parents and educators sometimes report high rates of refusals and stubbornness (Kasari & Freeman, 2001) and this in conjunction with poor task persistence may be associated with lower levels of achievement in the long term and thus should be a target of early intervention efforts (Fidler, 2006; Gilmore & Cuskelly, 2009, 2017). Other common challenges include oppositionality, impulsivity, tantrums, and self-injurious or aggressive behaviors.

Challenging behaviors can interfere with learning, daily living, and social relationships, may be unsafe for the individuals or for others, or can limit opportunities for inclusion in education and community programs. Many underlying factors likely contribute, including communication challenges, unusual sensory responses, poor emotional regulation/coping, and rigidity. Challenging behaviors may have several different functions, including to gain social attention, to obtain access to a tangible item or activity, to escape or avoid a situation, or to provide sensory or self-stimulation.

A medical evaluation is warranted in the setting of new or persistent behavioral challenges to assess for potential medical causes of challenging behaviors such as sources of pain, sleep disorders, hearing or vision impairment, celiac disease, thyroid disease, or gastrointestinal problems such as

constipation or reflux. It is also essential to consider psychosocial stressors or changes in home and school environments, as individuals with DS may be quite sensitive to even common life changes (Coe et al., 1999). An evaluation for co-occurring neurodevelopmental conditions (e.g., ASD or ADHD) or mental health conditions (e.g., depression, anxiety) should also be completed (see below), as the diagnosis may lead to specific, targeted behavioral interventions or pharmacological treatments.

Proactive, positive behavioral support strategies are the main treatment for behavioral challenges. These may include:

- Consistent use of routines
- Visual schedules, visual timers
- Social stories, video models
- Positive reinforcement of desired behaviors (praise, token economy reward systems)
- Distraction, redirection, or planned ignoring of undesired behaviors

Specific strategies will depend on the severity and frequency of behavior, presumed underlying function of the behavior, and the chronological age and developmental level of the child. Consistency in strategies and response to challenging behaviors across settings is essential, as is parent training on effective behavioral approaches. See Feeley and Jones (2006) for an in-depth review of literature and strategies to address challenging behaviors using the methodology of Applied Behavior Analysis (ABA) for assessment and intervention (Feeley & Jones, 2006).

Pharmacological treatment, when pursued, is driven by target symptoms that cause functional impairment (e.g., hyperactivity, impulsivity, mood issues, agitation/aggression), or by diagnosed co-occurring psychiatric conditions (e.g., ADHD, Depression, Anxiety, OCD). While some small studies of psychopharmacological treatments have been conducted in DS (Capone, Brecher, & Bay, 2016; Capone, Goyal, Grados, Smith, & Kammann, 2008, Roche et al., 2021), and published expert recommendations exist to guide pharmacological treatment approaches (Palumbo & McDougale, 2018), psychopharmacological treatments have not been adequately or systematically studied with gold-standard, rigorous approaches in DS. Typically, medications should be started at lower doses with slower escalation to avoid side effects. It is particularly important to consider side effect profiles and interactions with potential co-occurring conditions when choosing pharmacological treatments in DS. For example, cardiac structural and conduction problems should be considered, as should propensity for weight gain/overweight/obesity, constipation, or fatigue.



11. Autism spectrum disorder

Autism spectrum disorder (ASD) is a neurodevelopmental disorder characterized by difficulties with social interaction and communication, and by restrictive and repetitive patterns of behavior that cause impairment in learning and functions in daily life ([Diagnostic and Statistical Manual of Mental Disorders: Dsm-5, 2013](#)). The term “spectrum” refers to the wide range of symptom constellations and severity levels that vary greatly between individuals. ASD was previously considered rare in DS but has been increasingly recognized. Historically, cultural stereotyping has promoted the false notion that all people with DS are friendly, charming, and overtly social. This viewpoint can perpetuate a feeling of isolation for parents of children with DS who do not “fit” this oversimplified characterization, especially those with co-occurring ASD. Studies report varying prevalence, ranging from 7% to 42% depending on how the children were diagnosed and study methodology ([DiGuiseppi et al., 2010](#); [Lowenthal, Paula, Schwartzman, Brunoni, & Mercadante, 2007](#); [Oxelgren et al., 2017](#)). A meta-analysis of ASD rates in genetic syndromes reported ASD in 16% of people with DS (1 in 6) ([Richards, Jones, Groves, Moss, & Oliver, 2015](#)). Given that ASD is diagnosed in about 1/44 children (2.3%) in the general population ([Maenner et al., 2021](#)), this represents a substantially increased risk of ASD for children with DS.

When diagnosed through a careful evaluation by experienced specialists, ASD may be reliably diagnosed by 18–24 months in the general population ([Zwaigenbaum et al., 2015](#)), although most individuals are not diagnosed until later in childhood. Among children with DS, reliable diagnosis of ASD typically comes at an even older chronological age than in the general population for a variety of reasons ([Reilly, 2009](#)). Although ASD and ID can co-exist, an accurate diagnosis of ASD requires that social communication abilities be discrepantly lower than expected relative to the individual’s overall level of development or cognition. Thus, the diagnosis of ASD in DS can be challenging ([Gray, Ansell, Baird, & Parr, 2011](#)), and clinicians must differentiate ASD symptoms from the broader phenotype associated with DS.

In clinical practice, ASD may be over-diagnosed in DS because of the natural overlap in symptoms between ID and ASD and because of the coupling of specific therapeutic services with particular diagnoses (i.e., Applied Behavioral Analysis (ABA) therapy and ASD). However, ASD is also likely under-diagnosed at times due to diagnostic overshadowing and the lack of

standardized measures normed for diagnosis of ASD among children with DS. Additionally, medical conditions that are prevalent in DS may mimic or precipitate certain symptoms that could be confused with ASD. For these reasons, a comprehensive evaluation by an expert who has experience with both DS and ASD is recommended to make an accurate diagnosis of ASD in individuals with DS.

Formal evaluation with standardized assessments of cognition, language level, adaptive skills, social communication abilities, and restrictive and repetitive behaviors improve diagnostic accuracy and can be critical in determining whether social communication is impaired relative to overall developmental abilities. Some screening tools such as the m-CHAT, SRS, and SCQ have been studied in DS, as have diagnostic tools such as the ADOS and ADI-R (DiGuseppi et al., 2010; Hepburn, Philofsky, Fidler, & Rogers, 2008; Magyar, Pandolfi, & Dill, 2012; Oxelgren et al., 2017). However, further research is needed to understand the best use of these assessments and their interpretation in DS.

Research assessing symptom profiles in children with DS and co-occurring ASD compared to those with DS alone reveal that, as a group, those with DS and co-occurring ASD tend to have more significant impairments in cognitive abilities and adaptive skills, more difficulty with receptive and expressive language, more complex and interfering repetitive behaviors, and more mood and behavioral challenges (Capone, Grados, Kaufmann, Bernad-Ripoll, & Jewell, 2005; Carter et al., 2007; Channell et al., 2019; Molloy et al., 2009). Conversely, when compared to children with non-syndromic ASD in the general population, those with DS and co-occurring ASD may display lower levels of social communication impairment and less substantial repetitive behavioral patterns (Godfrey et al., 2019).

Clinically, signs and symptoms that are typically reported more commonly in children with DS and ASD compared to DS alone include: few attempts to communicate with others, decreased or unusual eye contact, decreased use of gestures and protodeclarative pointing, decreased social interest or social relatedness, more social withdrawal and decreased interest in others, more dysregulation with extreme meltdowns and difficulty being soothed, unusual or repetitive play such as looking at things from strange angles, dangling objects, licking objects, and more extreme sensory sensitivities. About half of children with DS and ASD present with developmental regression of social communication and language skills, although the regression is usually observed at a later age in those with DS, compared to age of

regression typically seen in ASD (Castillo et al., 2008). Symptoms of ASD are variable among individuals, and not all children with DS and ASD will exhibit all symptoms of social dysfunction and restricted and repetitive behavior.

More research is needed to better understand risk factors for ASD in DS. Some research has indicated that male gender, lower IQ, and certain medical conditions such as infantile spasms are likely to be risk factors for autism in DS (Molloy et al., 2009; Rasmussen, Börjesson, Wentz, & Gillberg, 2001), but other studies have not found consistent results. For example, one study recently found a female preponderance in the group with highest ASD symptom profiles (Fidler et al., 2022). Some research has also shown that a family history of ASD traits, communication/speech problems, learning difficulty, intellectual disability, or severe attention deficit may increase the risk of ASD in DS (Ghaziuddin, 2000; Rasmussen et al., 2001).

The mainstay of treatment for autism spectrum disorder are educational and behavioral therapies, such as Applied Behavioral Analysis (ABA). ABA is a specific therapeutic approach that involves breaking down complex skills into small steps and explicitly teaching those steps using modeling, repetition, and reinforcement, which can be particularly beneficial in DS. It is a systematic way of teaching, and usually involves direct 1:1 or 1:2 instruction with consistent reinforcement. ABA can be an effective strategy for increasing language, behavior, social, and self-care skills and for reducing challenging and maladaptive behaviors that interfere with daily life and learning in children with neurodevelopmental disorders, including both ASD and ID without accompanying ASD. There are many different models and types of behavioral therapy, some of which are more naturalistic and play based. Individuals with DS and ASD also may benefit from behavioral supports such as visual schedules, video modeling, and social stories as well as sensory-based supports.

There are no effective medications for treatment of core symptoms of ASD. However, because there are often accompanying co-occurring psychiatric symptoms, medications may be used in individuals with DS and ASD to target anxiety, depression, obsessive compulsive symptoms, irritability, aggression, self-injurious behaviors, impulsivity, hyperactivity, inattention, and sleep dysfunction (Palumbo & McDougle, 2018).



12. Attention deficit hyperactivity disorder

Attention Deficit Hyperactivity Disorder (ADHD) is one of the most common neurodevelopmental disorders of childhood, occurring in 9.4% of

the general population ([Data and Statistics About ADHD | CDC, 2022](#)). ADHD can be characterized as predominantly inattentive, predominantly hyperactive/impulsive, or as a combination of both ([Diagnostic and Statistical Manual of Mental Disorders: Dsm-5, 2013](#)). Children with DS may have higher risk for ADHD, with some prevalence reports ranging from 10% to 44% ([Edvardson et al., 2014](#); [Ekstein, Glick, Weill, Kay, & Berger, 2011](#); [Oxelgren et al., 2017](#)). Research has shown prematurity to be a risk factor for ADHD in DS ([del Hoyo Soriano et al., 2020](#)). The impact of age and gender differences on ADHD in DS is yet to be determined ([Edvardson et al., 2014](#); [Esbensen, Vincent, et al., 2022](#); [Ornoy, Rihtman, & Parush, 2011](#)).

ADHD can be challenging to diagnose in DS, as many of the symptoms are also observed in children with DS alone, indicating they may be part of the behavioral phenotype typically seen in DS ([Esbensen, Epstein, et al., 2022](#)). Additionally, challenges with executive functioning, such as working memory, emotional control, inhibition, and planning, which are closely tied to inattention and impulsivity, are also frequently found in DS ([Daunhauer et al., 2014](#); [Tungate & Conners, 2021](#)). Medical conditions that commonly occur in DS, such as thyroid disease, celiac disease, sleep problems, hearing or vision problems, or seizures can impact attention and impulsivity. Additionally, a child may appear inattentive or may display challenging behaviors if the schoolwork, environment, or demands are not appropriately matched to the child's developmental level or needs. For these reasons, a comprehensive evaluation for ADHD is recommended.

Evaluation for co-occurring ADHD in DS should include a thorough medical and developmental history and physical examination. It is especially important to get information from parents and caregivers as well as from teachers or therapists who observe the child in different settings. Notably, variability in informant reports has been consistently described in DS ([Edvardson et al., 2014](#); [Esbensen, Epstein, et al., 2022](#); [Ornoy et al., 2011](#); [Oxelgren et al., 2017](#)), and may have implications for clinical decision making. It is essential to consider the child's overall developmental functioning when considering a diagnosis of ADHD, although a formal diagnosis should not preclude appropriate interventions including behavioral, environmental, and/or pharmacological strategies. Standardized developmental scales such as the Vanderbilt ADHD Rating Scales—Parent and Teacher Forms ([Bard, Wolraich, Neas, Doffing, & Beck, 2013](#); [Wolraich, Bard, Neas, Doffing, & Beck, 2013](#)) or the Conners Rating Scales, Third Edition (Conners-3) have been used to study ADHD in children and adolescents with DS, although sometimes the interpretation of the scales can be

challenging when used among children with developmental delays. Scales may be most helpful in establishing baseline functioning and monitoring effectiveness over time.

Treatments and intervention for ADHD in DS should include environmental modifications and accommodations, such as preferential seating, a daily schedule that breaks up larger tasks into smaller chunks, reward systems, staggering preferred and nonpreferred tasks, reminders and organizational assistance, and minimization of environmental distractions. Behavior therapy should be a part of ADHD symptom management, including caregiver training to help establish effective ways to respond to behavioral difficulties that are commonly seen in ADHD. Medication is one of the mainstays of treatment in ADHD, but there are very few studies investigating medication efficacy for ADHD symptoms in people with DS. Generally, ADHD is treated with stimulant medications (methylphenidates, amphetamines) which are available in short acting and long-acting forms, or non-stimulants such as alpha agonists (guanfacine, clonidine) or selective norepinephrine reuptake inhibitors (atomoxetine, viloxazine). Both guanfacine and methylphenidate have been studied in small numbers of children with DS and ADHD and have been found to be effective and well-tolerated (Capone et al., 2016; Roche et al., 2021), but more research is needed to guide clinical recommendations.



13. Sensory processing difficulties

Sensory processing involves the collection, integration, and interpretation of information from multiple sensory systems, including touch, movement, joint position, visual, auditory, smell, and taste. Sensory processing thereby influences an individual's responses to the demands of everyday life, impacting how an individual learns, moves, explores the environment, behaves, completes tasks, and interacts with others.

Sensory processing impairments tend to be common among individuals with DS and may contribute to challenging behaviors. Though most frequently described in ASD, sensory processing challenges can also occur in all individuals with DS, including those who do and do not have co-occurring ASD. Impairments may involve hypersensitivity or over-responsiveness (leading to aversion), hyposensitivity or under-responsiveness (leading to sensory seeking behaviors), difficulties with discrimination of different stimuli, and challenges with motor planning and proprioception (Camarata, Miller, & Wallace, 2020).

Prior research using the Sensory Profile, a scale used to assess sensory processing, has shown that parent report of low endurance with passive responses to stimuli are common among children with DS, in addition to under-responsiveness with sensory-seeking behaviors and trouble with auditory filtering (attending when multiple auditory stimuli are present) (Bruni, Cameron, Dua, & Noy, 2010; Wuang & Su, 2011). Under-responsive/sensory-seeking behaviors have also been shown to be a significant predictor of challenging behavior such as disruptive and self-directed behaviors (Will et al., 2019).

Common sensory related issues that have been described among individuals with DS include challenges with handling proprioceptive input, tactile aversions, and difficulties with auditory stimuli (Bruni, 2016). Differences in perceptions of proprioceptive input are thought to be related to hypotonia, leading to challenges with interpretation of sensory input from muscles and joints (Bruni, 2016). This may lead to a tendency toward rough play and crashing into things, seeking out deep pressure, chewing/biting behaviors, writing too hard, bumping into things, challenges going up and down stairs, and so forth. As such, increasing exposure to proprioceptive input may help the individual with DS to control and interpret body placement in space.

Tactile aversions tend to be common in DS, as some individuals may not like the feel of certain textures or touch on their hands or body or may not tolerate certain hygiene-related tasks due to sensory aversions, such as hair washing or haircuts, tooth brushing, or dressing. These aversions can negatively impact early exploration and learning as well as longer term adaptive skill development, so early messy play and other exposure techniques can be helpful in mitigating this tendency (Bruni, 2016).

Finally, individuals with DS may have difficulties with auditory filtration and in general experience more challenges with auditory processing. This may contribute to hypersensitivity to sudden loud or unexpected sounds or may cause the individual to become easily overwhelmed by crowded or noisy environments. At the same time preference for listening to favored auditory stimuli (usually music) at a loud volume may be seen, often confusing caretakers. Hearing loss and cerumen impaction can contribute to these auditory sensory differences and should be addressed.

Feeding is a common challenge among young children with DS, often related to sensory issues. This may start in infancy when food refusal or aversion may sometimes be a sign of dysphagia, potentially meriting an in-depth swallowing evaluation. Oral aversions may be seen more commonly among infants who undergo complex surgical procedures and are tube-fed during

infancy without introduction of solid foods at a time when the child is developmentally ready to work on oral-motor skill development (Anil, Shabnam, & Narayanan, 2019). Early signs of oral sensory issues include swallowing of food that is not chewed well enough, overstuffing or pocketing food, gagging, excessive drooling, mouthing of non-food items in older children, and intolerance of smells, sights, textures, temperatures, and colors of foods (Anil et al., 2019; Bruni et al., 2019). Working with a knowledgeable feeding therapist can be essential in these cases, with use of techniques to optimize opportunities for sensory and exploratory play, use of face and mouth massage and oral vibration, introduction of various food textures, temperatures, and strong flavors, and so forth (Bruni, 2016).

Sensory processing difficulties may improve throughout childhood or may persist to some degree throughout life. Individuals may benefit from ongoing therapeutic efforts to mitigate the onset of associated maladaptive behaviors and to promote self-regulation and participation in activities of everyday life to the most complete extent possible (Bruni, 2016). Working with an occupational therapist who is experienced with sensory processing can be extremely beneficial. A common approach is to utilize a “sensory diet,” which involves creation of a scheduled sensory activity program designed to target an individual’s specific needs and help to optimize function. A sensory diet should be implemented proactively and regularly throughout the day with the aim of reducing sensory and emotional overload by satisfying the individual’s sensory needs, thereby preventing downstream challenging behaviors.



14. Co-occurring mental health conditions

Individuals with DS may experience a variety of mental health symptoms at a higher rate than the general population (Rivelli et al., 2022). While challenging behaviors and other externalizing symptoms are common in younger children (as previously described) and may persist, the onset of mental health disorders with internalizing symptoms typically emerges in late childhood and adolescence (Capone et al., 2006; Dykens, 2007; Rice et al., 2015; Siegel & Smith, 2010). These include anxiety, OCD, and mood disorders. Psychosis may be seen rarely, and regression is a phenomenon that is increasingly recognized.

14.1 Anxiety disorders

Anxiety disorders tend to have an onset in late childhood and adolescence in DS. Symptoms may include excessive worrying, increasing situational

anxiety with transitions or in anticipation of new situations leading to avoidance (including school avoidance), hypervigilance or restlessness, and increases in rigidity and ritualistic behaviors like skin-picking and nail-biting. Individuals with DS may also manifest physical complaints such as stomachaches, headaches, or pacing. Specific phobias, agoraphobia, selective mutism, and panic disorders may also be seen. Assessment of symptoms and impact on function may rely heavily on caregiver and/or teacher report and behavioral observations including the use of standardized measures such as the SCARED (Screen for Child Anxiety Related Disorders) inventory and the BASC-3. Additionally, some patient-completed standardized questionnaires (SCARED-child) have been studied among adolescents with DS with a mild level of cognitive impairment and may be informative (Esbensen et al., 2003; Sánchez-Teruel & Robles-Bello, 2019).

Generally, management should include ruling out contributing medical conditions and initiation of therapy, such as modified cognitive-behavioral therapy. Use of relaxation techniques, coping strategies, desensitization approaches, and sensory supports also play a role in treatment and help the individual to build a skill set that they may recruit over time to deal with the onset of new anxious symptoms. Pharmacologic treatments can be highly effective in anxiety management, including use of SSRIs (e.g., sertraline, fluoxetine) and buspirone (Palumbo & McDougle, 2018; Wehry, Beesdo-Baum, Hennelly, Connolly, & Strawn, 2015).

14.2 Obsessive-compulsive disorder

Obsessive-compulsive disorder (OCD) is a specific anxiety disorder characterized by recurrent, unwanted, preoccupying thoughts (obsessions) and repetitive behaviors (compulsions) that one feels compelled to perform which significantly interfere with daily activities and social interactions. However, OCD needs to be differentiated from behavioral tendencies commonly seen in DS. In general, routinized and compulsive-like behaviors are quite common among individuals with DS, especially among younger children, with more frequent and intense engagement in these behaviors compared to the general population and populations with other forms of ID (Evans & Gray, 2000; Määttä, Tervo-Määttä, Taanila, Kaski, & Iivanainen, 2006). Additionally, many adults tend to prefer sameness, repetition, and order in their lives.

Experts often use the term “grooves” to describe these patterns and routines that individuals with DS tend to follow (McGuire & Chicoine, 2021). The function of grooves may be to help the individual organize and manage

their lives, cope with stress, exert control over their environment, and express preferences. Grooves can present a challenge, however, when an individual gets overly stuck in these patterns. Sometimes grooves or other compulsive behaviors escalate in pervasiveness and frequency to the point that they impair daily functioning, and, in these circumstances, OCD should be considered. The most obvious and interfering manifestation of OCD in DS beyond excessive inflexibility with routines tends to be compulsive behaviors like repeated ordering, arranging, checking, touching, hoarding, and handwashing/maintaining cleanliness. The individual may describe perseverative thoughts, but typically the obsessional component of OCD is more difficult to assess in DS.

Treatment approaches include behavioral strategies, modified cognitive-behavioral therapies including modified Exposure and Response Prevention (ERP) therapy, and pharmacologic intervention. Medications include SSRIs (often at higher doses), SNRIs, and augmentation with antipsychotic medication if OCD behaviors are extremely excessive and begin to take on a psychotic nature (Palumbo & McDougale, 2018; Sutor, Hansen, & Black, 2006).

14.3 Depression

Depression may be difficult to detect among individuals with DS, especially those who are less verbal or who have more significant levels of ID. Depression is associated with sad or low mood, though this may not be self-reported by individuals with DS. Instead, depression may manifest as social withdrawal, decreased interest in preferred activities, and notable changes in sleep, appetite, activity level, and engagement in self-care activities (Capone et al., 2006; Walker, Dosen, Buitelaar, & Janzing, 2011). Medical causes for such changes should be ruled out first, including OSA, hearing or vision changes, unidentified sources of pain (musculoskeletal, constipation, etc.), hypothyroidism, and celiac disease. In DS, there tends to be an association between the onset of depressive symptoms and psychosocial stressors that may be perceived as relatively minor to others (sibling moving to college, absence of a teacher, chronic illness in the family, death of a pet, etc.) (Dykens, 2007). Anticipation of potentially triggering events and provision of supportive counseling can help mitigate the impact of stressors. Treatment includes therapy and individual supports in combination with pharmacological treatment, typically with SSRIs (Dykens, 2007; Palumbo & McDougale, 2018; Walker et al., 2011).

14.4 Psychosis

Psychosis in DS is much less common than anxiety or mood related conditions. Hallucinations or true response to external stimuli should always be distinguished from self-talk, which many teens and adults engage in as a form of self-assurance or cognitive processing. While self-talk is common in DS, increasing negative self-talk may also be a manifestation of low mood, anxiety, or adjustment disorders (McGuire & Chicoine, 2021).

14.5 Unexplained regression in Down syndrome

There is an increasingly recognized phenomenon of unexplained regression occurring predominantly in adolescence, sometimes referred to as Down syndrome disintegrative disorder (DSDD), Down syndrome regression disorder (DSRD), or Unexplained Regression in Down syndrome (URDS). The age of onset of this condition makes it distinct from regressive ASD, which is a neurodevelopmental disorder with much earlier onset, and from Alzheimer's disease which is associated with neurodegeneration and regression in older adults. Previously acquired skills are lost, resulting in a severe functional deterioration that greatly impacts quality of life, participation in school and recreation activities, and family functioning. This may be a precipitous decline or a more gradual deterioration of functioning over time. Manifestations of this regression may include loss of language or change in sentence structure and prosody (such as using a whisper voice), social withdrawal and retreat into internal worlds, personality changes, decline in ability to independently carry out activities of daily living (e.g., toileting, feeding, bathing, and dressing), sleep disturbance, new behavior problems, or psychosis and catatonia (Ghaziuddin, Nassiri, & Miles, 2015; Miles, Takahashi, Muckerman, Nowell, & Ithman, 2019; Santoro et al., 2020; Walpert, Zaman, & Holland, 2021). Motor changes such as stiffness, slow or freezing movements, and/or an increase in stereotypies and abnormal movements are often described (Ghaziuddin et al., 2015; Miles et al., 2019; Santoro et al., 2020; Walpert et al., 2021). This may occur in the context of a recognizable psychiatric disorder such as depression, medical conditions such as sleep apnea or Hashimoto's disease (Mircher et al., 2017), an autoimmune/inflammatory process (Cardinale et al., 2019; Jacobs, Schwartz, McDougle, & Skotko, 2016), or may occur in response to environmental changes/life events or psychosocial stressors (Mircher et al., 2017; Stein, Munir, Karweck, Davidson, & Stein, 2013). Definitive explanations for regressive symptoms in DS are limited in current literature, despite most

patients undergoing extensive evaluation for neurological causes including seizures or structural brain anomalies, encephalitis, or other medical problems such as thyroid disease or OSA (Miles et al., 2019; Santoro et al., 2020). Multiple treatments have been used with variable success including psychopharmacological treatments (e.g., SSRIs, neuroleptics), directed treatments for catatonia (e.g., high dose benzodiazepines or electroconvulsive therapy—ECT), or immunotherapy (Hart, Worley, Kishnani, & van Mater, 2021; Rosso et al., 2020; Walpert et al., 2021). A return to baseline functioning is highly variable in individuals with a diagnosis of unexplained regression, and language tends to be the skill that is slowest to reemerge (Mircher et al., 2017).



15. Conclusion

Down syndrome, the leading genetic cause of intellectual disability, is associated with a distinct phenotype characterized by common physical features, co-occurring medical conditions affecting multiple body systems, and significant neurological and neurodevelopmental manifestations. Extensive neurobehavioral and cognitive phenotyping has revealed a characteristic neurobehavioral profile. Overall development is delayed in most domains, but with prominent delays in early motor and expressive language skills. Impairments in cognition range from mild to profound with a characteristically uneven profile of cognitive development and a unique profile of abilities. There are relative strengths in visual processing and nonverbal cognitive abilities compared to auditory processing and verbal cognitive abilities. Significant language and communication challenges are evident throughout the lifespan. There is also substantial inter-individual variability in cognition, language, and social abilities, though the underlying etiology for these differences remains largely unknown. The neurobiological mechanisms that underly these developmental and behavioral impairments are just beginning to be elucidated. Research on the pathophysiology of DS provides the foundation necessary for the development of future biological interventions which might restore these aberrant neurobiological processes.

There are multiple medical conditions that people with Down syndrome are at higher risk of developing which may impact brain development and subsequent functioning and behavior. However, it remains difficult to assess the neurodevelopmental impact of various health conditions in DS given the inter-individual variability in cognitive, language, and social abilities. For example, infantile spasms, the most common epilepsy syndrome in children

with Down syndrome, is associated with lower cognitive abilities and higher risk of autism spectrum disorder. While some studies of children with DS and infantile spasms have suggested factors that result in worse outcomes (e.g., delayed recognition and initiation of treatment), findings have been inconsistently replicated using very small sample sizes. Future studies that take advantage of multi-center collaborative efforts to increase sample size should be promoted to address this issue. Additionally, continued efforts to develop more standardized and valid outcome measures particular to this population may improve detection of true differences among individuals with DS and various co-occurring conditions.

Obstructive sleep apnea is another condition highly prevalent in Down syndrome that is likely to have developmental and behavioral consequences when not promptly recognized and treated. Research in both the general population and DS reveals higher rates of inattention, more impulsive and hyperactive behaviors, and lower overall developmental scores in those with more severe OSA. Additionally, parasomnias and behavioral sleep challenges can limit sleep efficiency. Diagnosis and treatment of sleep disturbance is important to maximize neurodevelopmental and behavioral functioning, though further research looking into various treatment strategies and subsequent outcomes is needed to inform best clinical care practices.

Several health conditions prominent in Down syndrome have been shown to have an impact on development in the general population, though the developmental impact of such conditions has not been adequately studied or consistently reported in Down syndrome. For example, congenital conditions requiring early surgical intervention and prolonged hospitalization (e.g., congenital heart disease, duodenal atresia, or tracheo-esophageal fistulas), congenital hypothyroidism, severe iron deficiency anemia, and celiac disease have all been associated with developmental and behavioral impairments in the general population, but with varying reports or limited research in Down syndrome. Hearing and vision impairments are particularly common in children with Down syndrome and can impact learning and behavior, requiring routine screening and monitoring to prevent neurodevelopmental consequences and to optimize developmental outcomes.

Neurological conditions seen in DS include near universal early hypotonia and increased risk for seizures, movement disorders, stereotypies, and motor and vocal tics. Additionally, challenging behaviors are very common in children and adolescents with Down syndrome, including inflexible cognitive style, noncompliant behavior, task avoidance, or even aggressive or self-injurious behaviors. Sensory processing differences may also contribute to

behavioral challenges. Behavioral change warrants evaluation for potential medical causes including sources of pain (constipation, reflux, joint pain, etc.), sleep problems, hearing or vision changes, celiac disease, thyroid disease, and other considerations unique to the individual. Positive behavioral support strategies, environmental modifications, and dedicated behavioral therapy are often effective in managing behavioral challenges.

Neurodevelopmental disorders such as autism spectrum disorder or attention-deficit hyperactivity disorder can co-occur in DS with symptoms emerging in early childhood. The presentation of these conditions in Down syndrome can be complex, often with nuanced differences compared to children with these conditions who do not have DS. It is important to assess social communication, attention span, behavior, and other aspects of these conditions in the context of overall developmental functioning. Formal evaluation involving assessment of medical conditions and standardized assessment of cognition, language, adaptive skills, social communication, and restrictive and repetitive behaviors can inform the diagnostic process and clarify relative strengths and weaknesses that are pertinent to intervention planning. More research is needed to better understand risk factors and best treatments and interventions for co-occurring neurodevelopmental conditions in DS.

Mental health conditions, such as depression, anxiety, and obsessive compulsive disorder can also occur, typically presenting during or after puberty. There is also a newly recognized phenomenon of unexplained regression in Down syndrome that presents with a precipitous decline in adaptive functioning and social communication abilities in addition to the onset of new behaviors, but is distinct from childhood-onset autism or early-onset Alzheimer's disease. Therapy and pharmacological treatment are typically recommended for these conditions based on expert opinion and collections of anecdotal reports, as there is limited data obtained from clinical trials to guide recommendations for intervention at this time. Treatment approaches are largely extrapolated from research in the general population. Dedicated safety and efficacy studies of psychopharmacological treatments in Down syndrome are a crucial next step in determining the most effective targeted treatments.

Clinicians need to be aware of the variety of medical problems that may occur in individuals with Down syndrome to optimize health and functioning. Recognition and targeted treatment of co-occurring neurological, neurodevelopmental, behavioral, and mental health disorders is critical. Clinical

care from those with expertise in DS can be essential for the appropriate detection, diagnosis, and management of these conditions. Diagnostic overshadowing, in which symptoms of other neurodevelopmental conditions are erroneously attributed to DS and ID alone, has historically interfered with appropriate diagnosis and treatment of these conditions for people with Down syndrome. While increased recognition and research in recent years has advanced the field, further investigation is needed to establish a robust evidence base to comprehensively guide clinical care of patients with DS who have complex needs. A better understanding of the interplay between medical conditions and neurodevelopment is a high priority for future research. Additionally, further research to clarify factors contributing to the heterogeneity in Down syndrome is essential to improving our grasp of underlying neurobiological mechanisms of functioning that might guide future targeted treatments.

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